

CHANGING THE SUBJECT OF A FORMULA

Changing the subject of a formula involves moving letters or numbers (terms) from one side of the formula to the other, leaving a new term on its own. The way to do this depends on whether the term being moved is positive (+c), negative (-c), or whether it is part of a multiplication (bc) or division (b/c). When moving terms, whatever is done to one side of the formula needs to be done to the other.

Moving a positive term

$-c$ is brought in to the left of the equals sign

$-c$ is brought in to the right of the equals sign

$$A = b + c$$

To make b the subject, $+c$ needs to be moved to the other side of the equals sign.

Add $-c$ to both sides. To move $+c$, its opposite ($-c$) must first be added to both sides of the formula to keep it balanced.

$$A - c = b + c - c$$

$+c - c$ cancels out because $c - c = 0$

$$A - c = b + \cancel{+c} - \cancel{+c}$$

Simplify the formula by canceling out $-c$ and $+c$ on the right, leaving b by itself as the subject of the formula.

a formula must have a single symbol on one side of the equals sign

$$A - c = b$$

The formula can now be rearranged so that it reads $b = A - c$.

Moving a negative term

$+c$ is brought in to the left of the equals sign

$+c$ is brought in to the right of the equals sign

$$A = b - c$$

To make b the subject, $-c$ needs to be moved to the other side of the equals sign.

Add $+c$ to both sides. To move $-c$, its opposite ($+c$) must first be added to both sides of the formula to keep it balanced.

$$A + c = b - c + c$$

$-c + c$ cancels out because $c - c = 0$

$$A + c = b - \cancel{c} + \cancel{c}$$

Simplify the formula by canceling out $-c$ and $+c$ on the right, leaving b by itself as the subject of the formula.

a formula must have a single symbol on one side of the equals sign

$$A + c = b$$

The formula can now be rearranged so that it reads $b = A + c$.

Moving a term in a multiplication problem

bc means $b \times c$

$\div c$ (or $/c$) is brought in to the left of the equals sign

$\div c$ (or $/c$) is brought in to the right of the equals sign

$$A = bc$$

In this example, b is multiplied by c . To make b the subject, c needs to move to the other side.

Divide both sides by c . To move the c to the other side, you must do the opposite of multiplying, which is dividing.

$$\frac{A}{c} = \frac{bc}{c}$$

c/c cancels out because c/c equals 1

$$\frac{A}{c} = \frac{b\cancel{c}}{\cancel{c}}$$

Simplify the formula by canceling out c/c on the right, leaving b by itself as the subject of the formula.

a formula must have a single symbol on one side of the equals sign

$$\frac{A}{c} = b$$

The formula can now be rearranged so that it reads $b = A/c$.

Moving a term in a division problem

b/c means $b \div c$

$\times c$ is brought in to the left of the equals sign

$\times c$ is brought in to the right of the equals sign

$$A = \frac{b}{c}$$

In this example, b is divided by c . To make b the subject, c needs to move to the other side.

Multiply both sides by c . To move the c to the other side, you must do the opposite of dividing, which is multiplying.

$$A \times c = \frac{b \times c}{c}$$

c/c cancels out because c/c equals 1

$$A \times c = \frac{b\cancel{c}}{\cancel{c}}$$

Simplify the formula by canceling out c/c on the right, leaving b by itself as the subject of the formula.

remember that $A \times c$ is written as Ac

a formula must have a single symbol on one side of the equals sign

$$Ac = b$$

The formula can now be rearranged so that it reads $b = Ac$.